c) $x^{(0)} = (1.1, 0.9, 1.00)^T$ find $x^{(1)}$

k =	x_1	x_2	x_3
0	1.1	0.9	1.00
1	1.0200	1.64999	

$$x_1^{(1)} = 0x_1^{(0)} + \left(\tfrac{-1}{5}\right)x_2^{(0)} + \left(\tfrac{-2}{5}\right)x_3^{(0)} + \frac{8}{5}$$

$$= 0 - (0.20000)(0.90000) - (0.40000)(1.0000) + 1.60000$$

$$= 1.02000$$

$$x_2^{(1)} = \left(\tfrac{-1}{3}\right)x_1^{(0)} + 0 + \left(\tfrac{1}{3}\right)x_3^{(0)} + 1$$

$$= (-0.33333)(1.10000) + 0 + (0.33333)(1.00000) + 1$$

$$= 0.96666$$

$$x_3^{(1)} = \left(\tfrac{-2}{5}\right)x_1^{(0)} + \left(\tfrac{-2}{5}\right)x_2^{(0)} + 0 + \tfrac{9}{5}$$

$$= (-0.40000)(1.10000) + (-0.40000)(0.90000) + (1.80000)$$

$$= 1.0000$$

+4

$$x^{(1)} = \begin{bmatrix} 1.0200 \\ 0.96666 \\ 1.0000 \end{bmatrix}$$